The Joule-Thompson effect describes why allowing nitrous oxide to escape the gas cylinder at high flows can cause the tank to accumulate frost on the cylinder outlet. When a gas is allowed to exit its cylinder rapidly, the pressure inside the tank drops. If the pressure drops, then the amount of energy per area decreases as the gas expands. The decrease in energy results in a lowered temperature. This is referred to as the adiabatic expansion of a gas. The opposite is true also, if you compress a gas rapidly, the energy of the compressed gas increases and the temperature rises.

Not all gas expansion scenarios are considered adiabatic processes, however. If you place a patient on nasal oxygen at 2L/min using an oxygen cylinder, the oxygen tank will not form frost on it. The expansion of oxygen within the tank occurs slow enough at lower flow rates that the temperature of the gas equilibrates with the ambient environment and the cooling effect is less evident. At a low flow rate, the expansion of the gas is not considered an adiabatic process.

Nagelhout JJ, Plaus K. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 225.

Copyright 2015 Prodigy Anesthesia, Inc. It is unlawful to copy, distribute, and/or sell this material.